

# Longest cycles meet linearly in highly connected graphs

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# Overview

- Scott Smith's conjecture on longest cycles
- Background and previous ideas
- Our idea and three ingredients

# Smith's conjecture

This work is motivated by the following conjecture.

## Conjecture (Scott Smith, 1979)

*For every integer  $k \geq 2$ , any two longest cycles in a  $k$ -connected graph share at least  $k$  vertices.*

**Sharp example.** In  $K_{k,n-k}$ , where  $n \geq 3k$ , two longest cycles can intersect in exactly the  $k$  vertices of the smaller part.

Throughout the talk, connectivity means vertex-connectivity.

Grötschel (1984) attributes the conjecture to Scott Smith.

# Smith's conjecture



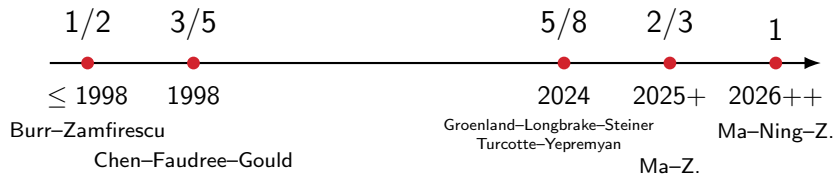
- $k \leq 6$ : Grötschel (1984);  $k = 7, 8$ : Stewart–Thompson (1995).
- $k \geq (n+16)/7$ : Gutiérrez–Valqui (2024).

# A quantitative problem

## Quantitative form

Let  $f(k)$  be the largest universal lower bound on the intersection of two longest cycles in a  $k$ -connected graph.

exponent of  $k$  in  $f(k)$



# Main theorem

## Theorem (Ma–Ning–Z., 2026++)

*For every integer  $k \geq 2$ , any two longest cycles in a  $k$ -connected graph share at least  $k/4000$  vertices.*

- This is the first linear lower bound in  $k$ .
- The order of magnitude is sharp; the constant is not optimized.

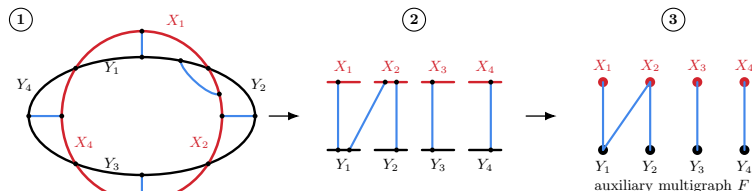
# The previous idea

Let  $X$  and  $Y$  be the two longest cycles under consideration. Set

$$M = V(X) \cap V(Y), \quad m = |M|.$$

Each of  $X - M$  and  $Y - M$  consists of at most  $m$  subpaths.

$k$ -connectivity gives  $k - m$  disjoint  $(X - M, Y - M)$ -paths in  $G - M$ .

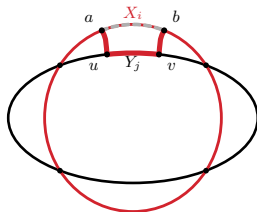


red/black subpath  $\mapsto$  vertex of  $F$ ;      blue path  $\mapsto$  edge of  $F$ .

# The previous idea

## Fact

*The auxiliary multigraph  $F$  has no parallel edges.*



two parallel edges  $\implies$  a longer cycle.

**(Burr–Zamfirescu)**

$$k - m \leq e(F) \leq m^2 \implies m \geq \sqrt{k} - 1.$$

## The previous idea

Suppose a graph  $J$  is forbidden in  $F$  and

$$\text{ex}(N, J) = O(N^\alpha).$$

Since  $|V(F)| \leq 2m$  and  $e(F) \geq k - m$ ,

$$k - m \leq e(F) = O(m^\alpha) \quad \implies \quad m = \Omega(k^{1/\alpha}).$$

### Observation

A forbidden structure in  $F$  with a smaller Turán exponent gives a better exponent on  $k$ .

# The previous idea

| Authors                   | Forbidden structures            | Resulting scale |
|---------------------------|---------------------------------|-----------------|
| Chen–Faudree–Gould (1998) | $K_{3,257}$ -free               | $k^{3/5}$       |
| Groenland et al. (2024)   | $Q_3^+$ -free                   | $k^{5/8}$       |
| Ma–Zhao (2025)            | $O( V(F) )$ copies of $K_{2,7}$ | $k^{2/3}$       |

# Current approach

## Theorem (Ma–Ning–Z., 2026++)

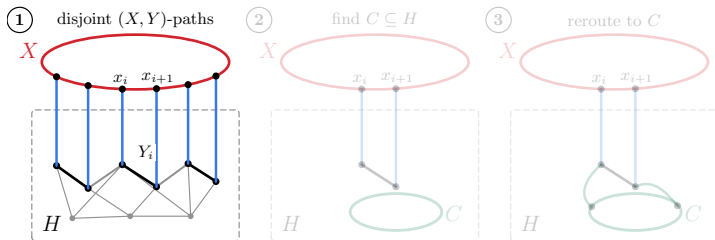
*For every integer  $k \geq 2$ , any two longest cycles in a  $k$ -connected graph share at least  $k/4000$  vertices.*

The following special case conveys the main proof idea.

## Model case

Let  $X, Y$  be longest cycles in a  $k$ -connected graph and  $m = |V(X) \cap V(Y)|$ . Suppose that every component of  $G - X$  is 3-connected and contains  $k$  vertices. Then  $m = \Omega(k)$ .

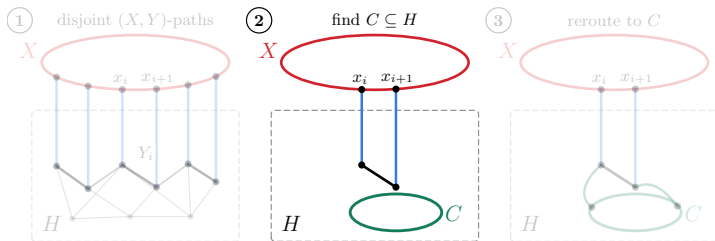
# Proof idea in the model case



Choose a component  $H$  of  $G - X$  containing a longest subpath of  $Y - X$ .

Menger's theorem gives many  $(X, Y)$ -paths ending in  $H$ .

# Proof idea in the model case



Let  $p(H)$  and  $c(H)$  denote the lengths of a longest path and a longest cycle in  $H$ .

- ④ The  $m$  subpaths of  $Y - X$  contain at least  $c(G) - 2m$  edges, so

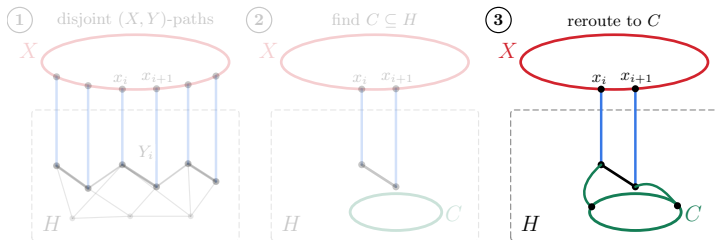
$$p(H) \geq \max_{i \in [m]} |Y_i| \geq \frac{c(G)}{m} - 2.$$

Hence  $H$  contains a cycle  $C$  with  $|C| = \Omega(c(G)/m)$ .

## Theorem (Bondy–Locke)

Every 3-connected graph  $H$  satisfies  $c(H) \geq \frac{2}{5}p(H)$ .

# Proof idea in the model case



- ② Take two paths with consecutive endpoints  $x_i, x_{i+1}$  on  $X$ .  
By 2-connectivity, there are disjoint  $(X, C)$ -paths in  $G$ . Hence,

$$\text{dist}_X(x_i, x_{i+1}) \geq |C|/2.$$

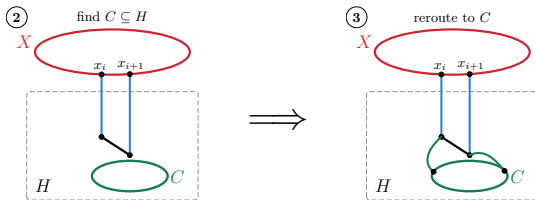
- ③ Summing over all  $k$  consecutive pairs,

$$c(G) \geq k \Omega\left(\frac{c(G)}{m}\right) \implies m = \Omega(k).$$

# Two key properties in the proof

## Property 1

For some suitable cycle  $C \subseteq H$ , any two disjoint  $(X, H)$ -paths can convert to two disjoint  $(X, C)$ -paths.



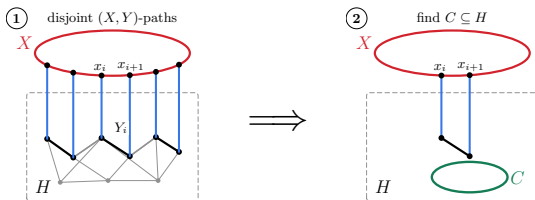
This is a weak form of 2-connectivity.

# Two key properties in the proof

## Property 2

Given a path  $P$  in  $H$ , find a cycle  $C \subseteq H$  with

$$|C| = \Omega(|P|).$$



Bondy-Locke's theorem guarantees this in 3-connected graphs.

# Roadmap of the new proof

## Achieve Property 1

Cover all edges of  $Y - X$  by  $O(m)$  subgraphs, each satisfying a weak form of 2-connectivity.

## Achieve Property 2

Cover all but except for  $O(mk)$  edges of  $Y - X$  by  $O(m)$  paths, each of them has a corresponding cycle as in Property 2.

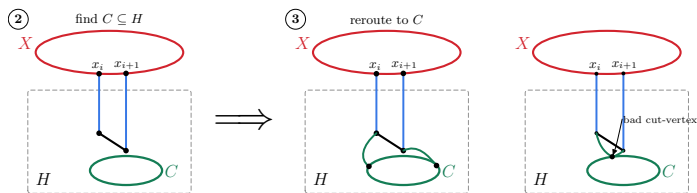
## The remaining range

In the range  $c(G) = O(mk)$ , use a strengthened Dirac-type theorem.

# Bad cut-vertex

## Property 1

For some suitable cycle  $C \subseteq H$ , any two disjoint  $(X, H)$ -paths can convert to two disjoint  $(X, C)$ -paths.



A bad cut-vertex in  $H$  is a cut-vertex that separates two specific paths.

Weak form of 2-connectivity:  $H$  contains no bad cut-vertex.

# The covering theorem

## Theorem (Weak 2-connectivity covering)

*There exists a collection of subgraphs  $\mathcal{H}$  of  $G - X$  such that  $E(\mathcal{H}) \supseteq E(Y - X)$ ,  $|\mathcal{H}| \leq 6m$ , and every  $H \in \mathcal{H}$  has no bad cut-vertex. Moreover, for every  $H \in \mathcal{H}$  and feasible cycle  $C \subseteq H$ , if  $x_1, x_2 \in X$  are the endpoints on  $X$  of two disjoint  $(X, H)$ -paths, then*

$$\text{dist}_X(x_1, x_2) \geq |C|/2.$$

This achieves Property 1.

## Comparison with block-cut decomposition

Block-cut decomposition gives stronger connectivity but the number of blocks is unbounded.

# Amplifiers

We now turn to Property 2.

## Property 2

Given a path  $P$  in  $H$ , find a cycle  $C \subseteq H$  with  $|C| = \Omega(|P|)$ .

## Definition $((H, \alpha)$ -amplifier)

A path  $P \subseteq H$  is an  $(H, \alpha)$ -amplifier if some cycle  $C \subseteq H$  satisfies

$$|E(C) \cap E(P)| \geq \alpha|P|.$$

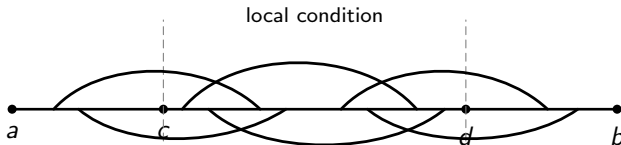
# Amplifiers

## Local amplifier lemma

Let  $P$  be an  $(a, b)$ -path in  $H$  and  $a <_P c <_P d <_P b$ . Assume:

- both  $P[a, c]$  and  $P[d, b]$  contain at least three vertices; and
- for every  $v \in V(P[c, d])$ , there are two disjoint  $(P[a, v], P[v, b])$ -paths.

Then  $P[c, d]$  is an  $(H, 1/6)$ -amplifier.



# Cover the paths by amplifiers and leftovers

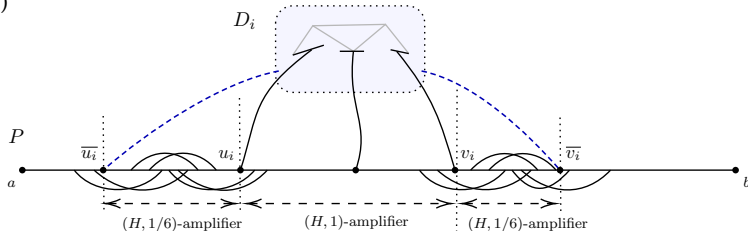
## Path covering lemma

There are three collections of subpaths of  $Y - X$  with totally  $O(m)$  paths, named  $\mathcal{U}, \mathcal{V}, \mathcal{W}$ , whose union covers  $E(Y - X)$ :

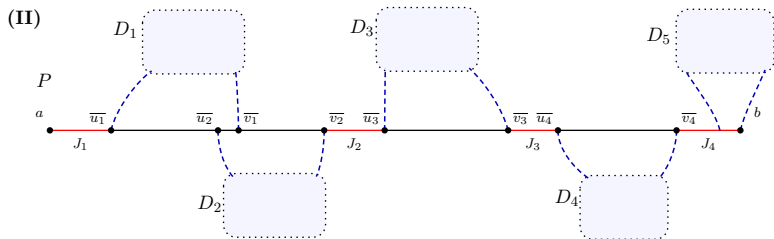
- $\mathcal{U}$  : 1/6-amplifiers,
- $\mathcal{V}$  : separated from all other segments by one  $O(m)$ -vertex set,
- $\mathcal{W}$  : paths of length at most four.

Each component  $D_i$  of  $H - P$  gives an interval on  $P$  covered by at most three amplifiers.

(I)



# Cover the paths by amplifiers and leftovers



The uncovered intervals have two controlled forms:

- one set of  $O(m)$  vertices separates them from all other segments; or
- they have length at most four.

## The remaining range

When  $c(G) \leq 100mk$ , this mechanism may fail: the total length allowed for the leftovers is already of order  $mk$ .

In this range we use a different approach, as a consequence of a strengthened Dirac-type theorem.

# Dirac's theorem to Jung's conjecture

The following result is classical.

## Theorem (Dirac, 1952)

*Every 2-connected  $n$ -vertex graph with minimum degree  $\delta$  contains a cycle of length at least*

$$\min\{2\delta, n\}.$$

Motivated by this, Jung conjectured that higher connectivity should increase the coefficient.

## Conjecture (Jung)

*Let  $G$  be an  $r$ -connected graph with minimum degree  $\delta$ , and let  $X$  be a longest cycle. If  $G - X$  contains a path on at least  $r - 1$  vertices, then*

$$|X| \geq r(\delta - r + 2).$$

## Jung's conjecture for $\delta \geq 6r$

We proved Jung's conjecture when  $\delta(G) \geq 6r$ .

### Theorem (Ma-Ning-Z., 2026+)

*Let  $G$  be  $r$ -connected with  $\delta(G) \geq 6r$ , and let  $X$  be a longest cycle. If  $G - X$  contains a path on at least  $r - 1$  vertices, then*

$$|X| \geq r(\delta(G) - r + 2).$$

**Application.** Suppose  $c(G) \leq 100mk$  and  $k > 4000m$ . Set  $r = \left\lfloor \frac{c(G)}{600m} \right\rfloor$ . Then

$$\delta(G) \geq k \geq 6 \cdot \frac{c(G)}{1200m} \geq 6r,$$

and  $G - X$  contains a path on at least  $c(G)/m - 1 \geq r - 1$  vertices. Hence

$$c(G) \geq r(\delta(G) - r + 2) \geq \frac{c(G)}{1200m} \cdot \frac{5k}{6} > c(G),$$

a contradiction. Therefore  $m \geq k/4000$ .

Ma-Ning-Zhao, arXiv:2606.03696 (2026). The current paper gives a shorter self-contained argument for this range.

## Three independent ingredients

- We establish the first linear lower bound towards Scott Smith's conjecture.
- A weakly 2-connected graph-covering theorem achieves Property 1.
- A generalization of Bondy–Locke, together with a path-covering theorem, achieves Property 2.
- A strengthened Dirac-type theorem handles the remaining range.

Thank you!